

Two-user egocentric QA generation pipeline

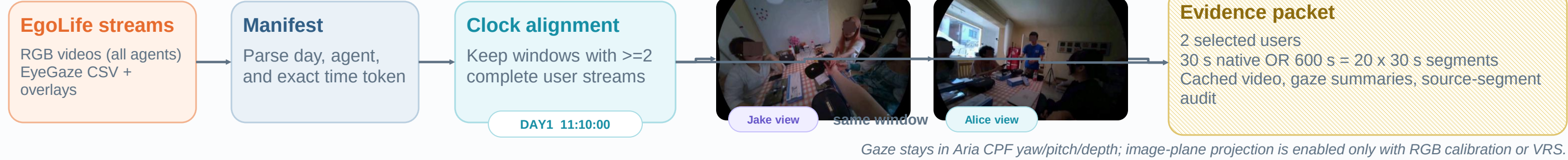
Synchronized streams -> CLIP-pruned generator media -> full-video verification -> auditable QA

BASELINE
PRODUCTION

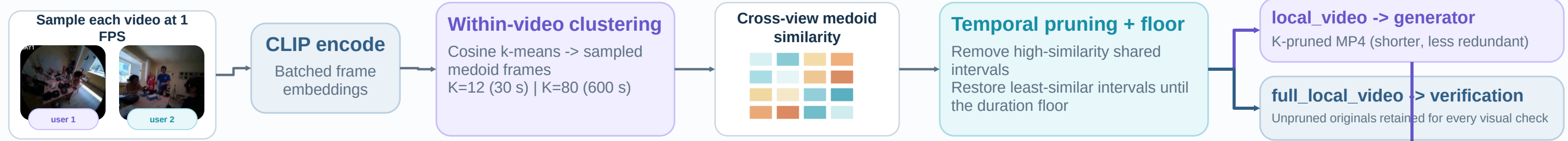
pruned media
full originals

retry feedback

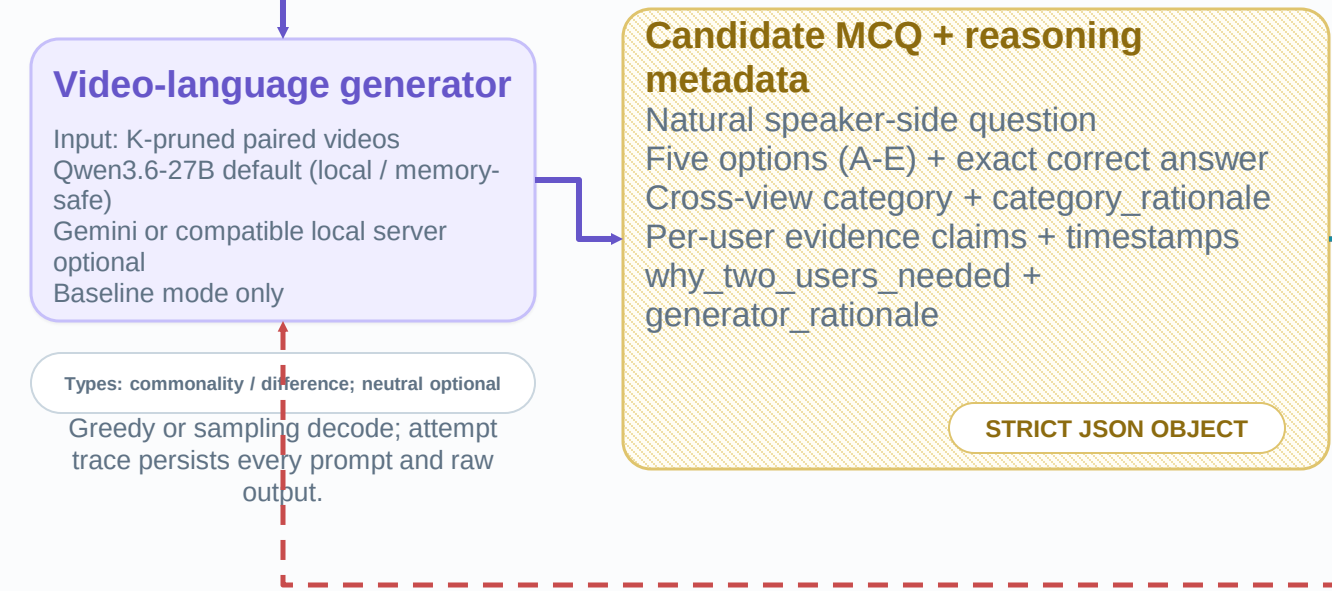
1 Build synchronized multi-user evidence



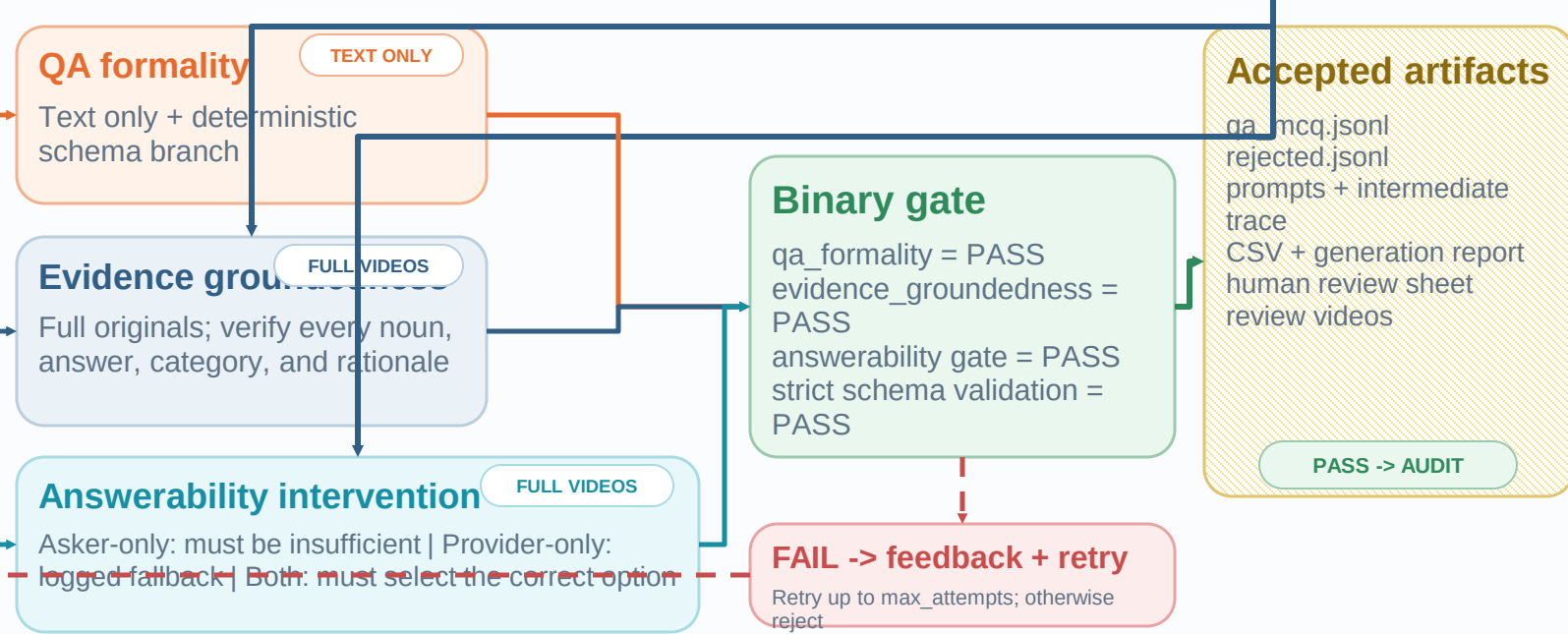
2 Prune redundant visual intervals and preserve full originals



3 Generate a situated two-user MCQ



4 Verify in parallel, gate, and retry



Audit and controlled experiments

Off the acceptance path: pruning K-grid with medoid/cluster traces + review.html | generator-rationale paired ablation | manual video review. Archived discovery / clip-guided modes and entropy scoring are not production gates.

Pruning illustration: what stays, what goes

Real fixed-pair K=12 run: DAY1 12:16:00, A4 Lucia + A6 Shure

K = 12

1 FPS

cosine >= 0.82

+/- 0.5 s interval

8 s keep floor

M1: 15 s <-> 26 s, cosine = 0.864

PRUNED

RESTORED

PRESERVED

VIDEO A - A4_LUCIA

Full original: 30.0 s
Sampled frames: 30
Pruned: 22.0 s
Generator keeps: 8.0 s

FULL ORIGINAL RETAINED

All 30 sampled frames

01234567891011121314151617181920212223242526272829

0s5s10s15s20s25s29s

Kept intervals: 0.0-1.5 | 4.5-7.5 | 15.5-16.5 | 27.5-30.0 s

8.0 s kept
22.0 s pruned

VIDEO B - A6_SHURE

Full original: 30.0 s
Sampled frames: 30
Pruned: 16.5 s
Generator keeps: 13.5 s

FULL ORIGINAL RETAINED

All 30 sampled frames

01234567891011121314151617181920212223242526272829

0s5s10s15s20s25s29s

Kept intervals: 0.5-8.5 | 11.5-13.5 | 17.5-20.5 | 29.5-30.0 s

13.5 s kept
16.5 s pruned

Generator route - local_video

Concatenate preserved + floor-restored intervals
Lucia: 8.0 s | Shure: 13.5 s
Temporal map links pruned time back to the originals

Verification route - full_local_video

Keep both unpruned 30 s originals unchanged
Evidence-groundedness + answerability always inspect full media
No pruned frame can hide an error from the judges

RESTORED = originally marked for removal, then reinserted from the least-similar candidates until the 8 s floor is met.